SAM ADEYEMI

Backend, Web & AI Software Engineer · Python / FastAPI · PHP / Drupal · LLMs, RAG & Agentic AI

SCAN FOR PORTFOLIO

- Sheffield, UK · eligible to work in the UK · open to hybrid
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6+ yrs
building and running production software

85%
faster workflow execution after re-engineering

73%
fewer output errors with schema-validated outputs

WCAG 2.2
AA audit run and all 10 issues fixed

PROFILE

Backend and AI software engineer with **6+ years** building secure, scalable services and shipping **LLM-powered features to production**, not just prototypes. Hands-on across the backend: REST APIs and integrations, PostgreSQL schema design and query optimisation, background jobs, real-time features, role-based access control and audit logging. On the AI side: OpenAI and Anthropic integration, structured outputs, RAG, **evaluation sets and guardrails**, and multi-agent orchestration. On the web side: around **15 years of WordPress**, and recent council-website builds in **Drupal 11 and LocalGov Drupal** with automated accessibility testing. MSc (Distinction), First Class BEng, and peer-reviewed AI governance research.

EXPERIENCE

Lead Software & AI Systems Engineer · Apex Digital Africa

2019 – Present

Web development and AI consultancy delivering backend, SaaS and AI products for clients and in-house platforms

- Design, build and maintain secure backend services, **REST APIs and third-party integrations** in Python and FastAPI, from requirements through testing, release and production support.
- Ship **LLM-powered features to production** using OpenAI and Claude APIs, with prompt design, structured outputs and context handling for retrieval and workflow automation.
- Re-engineered project workflows around **event-driven, metadata-driven design** with schema-validated outputs: **85% faster execution, 73% fewer output errors**.
- Design PostgreSQL schemas, **indexes and queries for large event datasets**, and build background jobs and real-time features that keep dashboards responsive.
- Implement **role-based access control, audit trails and reporting** for multi-user SaaS platforms, so every action is attributable.
- Deploy and run client software on **self-hosted Linux servers with Docker and Nginx**, backed by automated tests and CI pipelines.
- Build React and TypeScript front ends and WordPress sites; explain technical trade-offs to non-technical clients in plain English.

SKILLS

LANGUAGES
Python · PHP · SQL · TypeScript · JavaScript · C# (academic)

BACKEND & APIS
FastAPI · REST design · Integrations · Background jobs · Event-driven · Node.js / NestJS

AI & LLM
Claude API · OpenAI API · RAG · Evaluations · Guardrails · Cost controls · Multi-agent · Structured outputs

WEB & CMS
Drupal 11 · LocalGov Drupal · Twig · WordPress (15 yrs) · React · Tailwind

ACCESSIBILITY
WCAG 2.2 AA · GOV.UK patterns · axe-core · Playwright

DATA
PostgreSQL · MySQL · SQL Server · pgvector · Redis

QUALITY & DEVOPS
PHPUnit · pytest · GitHub Actions · Docker · Nginx · Linux · DDEV · Claude Code · Copilot

SECURITY
RBAC · Audit logging · Rate limiting · Secrets handling

SELECTED PROJECTS

Recent public-sector builds first. Each has automated tests that run on every change.

AI · PUBLIC SECTOR

Find My Service

github.com/SamAde1203/find-my-service

FastAPI assistant that answers residents' questions from a council's own pages using Claude, **citing every source**. Guardrails for emergencies, prompt injection and personal data; output checks on citations and links; no-AI fallback, daily cost cap and rate limiting.

✓ **53 evaluation questions in CI, including a held-out set · 44 tests**

Python · FastAPI · Claude API · BM25 · mypy · Docker

DRUPAL · ACCESSIBILITY

Demo City Council on LocalGov Drupal

github.com/SamAde1203/localgov-council-microsite

Council website on LocalGov Drupal 4 and Drupal 11. Custom module adds a **"Request an accessible format"** link to every PDF and Word document, with a request form, reference numbers and a staff queue. Ran a **WCAG 2.2 AA audit** and fixed all 10 issues found.

✓ **42 PHPUnit tests · 0 axe-core violations on 14 page states**

PHP 8.3 · LocalGov Drupal · Twig · Playwright · axe-core

DRUPAL · API INTEGRATION

Council Services on Drupal 11

github.com/SamAde1203/council-services-drupal

Custom Drupal 11 module and accessible theme: a postcode bin-day finder using the postcodes.io API, with caching, rate limiting and a fallback if the API is down.

✓ **26 PHPUnit tests · WCAG 2.2 AA checks in CI**

Drupal 11 · PHP · Form API · DDEV

AGENTIC AI · PRODUCTION

PHHM Multi-Agent AI Platform

Lead Architect & Developer

A central orchestrator routes work to five specialist agents (six in all) for communication, analytics, knowledge retrieval and member care, across website chat, WhatsApp, SMS and a voice line, serving users in several countries.

Python · FastAPI · OpenAI API · Linux

FINTECH SAAS · FOUNDER

SeedTracker

seedtracker.ng · Founder & Lead Engineer

Church finance platform live with 8 churches in Nigeria. Built backend services, payment and WhatsApp integrations, RBAC, audit trails and AI-assisted financial reporting.

React / TS · NestJS · FastAPI · PostgreSQL · Redis · Paystack

RAG · RESEARCH TOOLING

MR_CLAUDE_NOTEBOOK_LM

In development

Retrieval pipeline over technical documents (chunking, embeddings, vector storage and semantic search) behind a custom FastAPI backend, with answers validated against the retrieved evidence.

Python · FastAPI · pgvector · Docker

VALIDATION · CLINICAL RESEARCH

TrialForge AI & TDCT Validator

Rules-based validation engine running 20+ repeatable protocol-consistency checks, automating quality assurance and producing structured, auditable documentation. Python · YAML · MIT-licensed

EDUCATION

MSc Information Security & Software Engineering — Distinction
University of Greater Manchester

BEng (Hons) Software Engineering — First Class
University of Bolton

Built five data-driven web applications in C# / ASP.NET with SQL Server and MySQL during the degree.

RESEARCH

- Published peer-reviewed **AI governance research** in *AI and Ethics* (Springer Nature).
- **Peer reviewer** for the journal *Trials*.

HOW I WORK

- **Tests first, CI always:** every project here fails its build if quality drops.
- **Accessible by default:** GOV.UK patterns and WCAG 2.2 AA from the first commit.
- **Honest AI:** grounded answers, guardrails in code and measured with evaluations.

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